# Timestamp Migration – TvVTT & RadioClips

### Overview

After implementing 01_schema_update.sql. This migration adds a proper DATETIME column (TStamp_dt) to the TvVTT and RadioClips tables.

Previously, timestamps were stored as VARCHAR (TStamp), which made date comparisons slow and prevented efficient indexing.

### What This Change Does

- Adds a new column:
  - TStamp_dt DATETIME
- Converts existing TStamp values into proper DATETIME format
- Supports both formats currently found in the database:
  - YYYY-MM-DD HH:MM:SS
  - YYYY_MM_DD_HH-MM-SS
- Creates an index on TStamp_dt to enable efficient time-based queries

### Why This Was Needed

This change enables:

- Efficient 30–90 day retention deletes
- Faster recency-based search and filtering
- Indexed date comparisons (no full table scans)
- Improved scalability for large transcript tables

Without this change, retention logic required STR_TO_DATE() in queries, which prevents index usage and slows performance.

### Important Note for Future Inserts

This migration converts **existing data only**.

If new rows are inserted with only TStamp populated:

INSERT INTO TvVTT (TStamp) VALUES ('2026-03-01 12:00:00');

TStamp_dt will remain NULL unless:

- The application also populates TStamp_dt, or
- A database trigger is implemented

Future teams should ensure new inserts keep both fields consistent.

**Verification Queries**

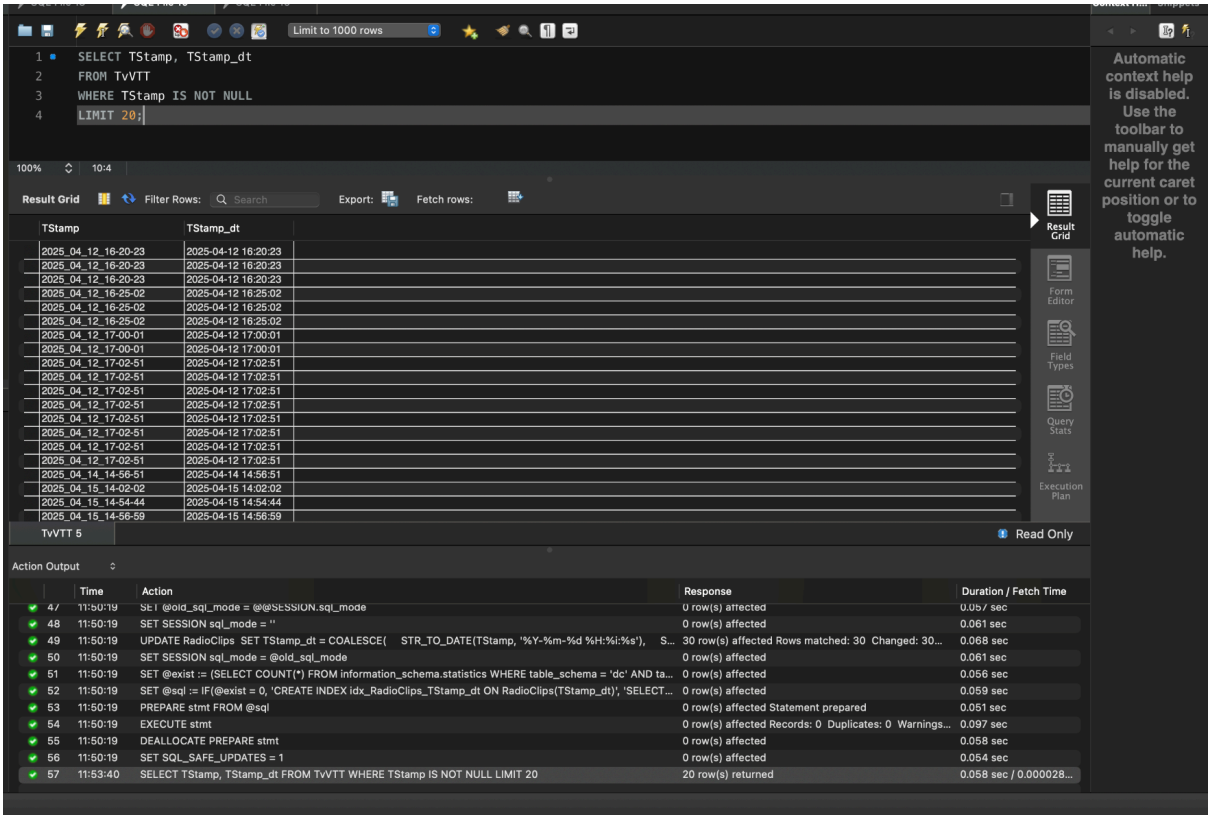
Check if all rows were converted:

```
SELECT COUNT(*)
FROM TvVTT
WHERE TStamp IS NOT NULL AND TStamp_dt IS NULL;
```

Confirm index usage:

```
EXPLAIN
SELECT *
FROM TvVTT
WHERE TStamp_dt < NOW() - INTERVAL 90 DAY;
```

The query should use the idx_TvVTT_TStamp_dt index.



# Retention Logic Audit – Existing Stored Procedure Verification

## Overview

Before implementing new retention procedures for TvVTT and RadioClips, existing stored procedures in the dc database were reviewed.

This was done to:

- Avoid duplicating existing functionality
- Prevent conflicts with legacy procedures
- Identify current retention logic
- Ensure the new TStamp_dt migration is properly utilized

## Procedures Reviewed

The following procedures were inspected:

```
SHOW PROCEDURE STATUS WHERE Db = 'dc';

SHOW CREATE PROCEDURE admin_del_radio;

SHOW CREATE PROCEDURE MediaRetentionCleanup;

SHOW CREATE PROCEDURE Insert_TvVTT;

SHOW CREATE PROCEDURE Insert_Radio;
```

## Findings

### 1. admin_del_radio(older_than_days)

- Deletes from RadioClips
- Uses DATEDIFF(NOW(), TStamp)
- Operates on TStamp (VARCHAR)
- Does not use TStamp_dt
- No logging mechanism

### 2. MediaRetentionCleanup()

- Deletes SRT records older than 90 days
- Attempts RadioClips cleanup using STR_TO_DATE(TStamp, ...)
- Uses function-based filtering (prevents index usage)
- Does not clean TvVTT
- Logs to media_retention_log, not cleanup_log

### 3. Insert_TvVTT / Insert_Radio

- Insert data into tables
- Only populate TStamp (VARCHAR)
- Do not populate TStamp_dt
- New rows may have TStamp_dt = NULL unless handled separately

**Summary**

Existing retention logic:

- Operates on VARCHAR timestamps
- Uses STR_TO_DATE() or implicit casting
- Prevents index usage
- Does not leverage the new TStamp_dt column
- Does not standardize logging into cleanup_log
- Does not clean TvVTT

No optimized DATETIME-based retention procedure currently exists for TvVTT.

Although legacy cleanup procedures exist, they do not utilize the newly introduced TStamp_dt column.

Therefore, implementing optimized retention procedures using indexed TStamp_dt is necessary and does not duplicate existing functionality.

**Verification Queries**

| | Time | Action | Response | Duration / Fetch Time |
|---|---|---|---|---|
| 44 | 11:50:18 | PREPARE stmt FROM @sql | 0 row(s) affected Statement prepared | 0.057 sec |
| 45 | 11:50:18 | EXECUTE stmt | 30 row(s) affected Records: 30 Duplicates: 0 Warnin... | 0.706 sec |
| 46 | 11:50:19 | DEALLOCATE PREPARE stmt | 0 row(s) affected | 0.059 sec |
| 47 | 11:50:19 | SET @old_sql_mode = @@SESSION.sql_mode | 0 row(s) affected | 0.057 sec |
| 48 | 11:50:19 | SET SESSION sql_mode = '' | 0 row(s) affected | 0.061 sec |
| 49 | 11:50:19 | UPDATE RadioClips SET TStamp_dt = COALESCE( STR_TO_DATE(TStamp, '%Y-%m-%d %H:%i:%s'), S... | 30 row(s) affected Rows matched: 30 Changed: 30... | 0.068 sec |
| 50 | 11:50:19 | SET SESSION sql_mode = @old_sql_mode | 0 row(s) affected | 0.061 sec |
| 51 | 11:50:19 | SET @exist := (SELECT COUNT(*) FROM information_schema.statistics WHERE table_schema = 'dc' AND ta... | 0 row(s) affected | 0.056 sec |
| 52 | 11:50:19 | SET @sql := IF(@exist = 0, 'CREATE INDEX idx_RadioClips_TStamp_dt ON RadioClips(TStamp_dt)', 'SELECT... | 0 row(s) affected | 0.059 sec |
| 53 | 11:50:19 | PREPARE stmt FROM @sql | 0 row(s) affected Statement prepared | 0.051 sec |
| 54 | 11:50:19 | EXECUTE stmt | 0 row(s) affected Records: 0 Duplicates: 0 Warnings... | 0.097 sec |
| 55 | 11:50:19 | DEALLOCATE PREPARE stmt | 0 row(s) affected | 0.058 sec |
| 56 | 11:50:19 | SET SQL_SAFE_UPDATES = 1 | 0 row(s) affected | 0.054 sec |
| 57 | 11:53:40 | SELECT TStamp, TStamp_dt FROM TvVTT WHERE TStamp IS NOT NULL LIMIT 20 | 20 row(s) returned | 0.058 sec / 0.000028... |
| 58 | 13:03:25 | SHOW TABLES LIKE '%cleanup%' | 1 row(s) returned | 0.061 sec / 0.000038... |
| 59 | 13:03:25 | SHOW TABLES LIKE '%log%' | 7 row(s) returned | 0.069 sec / 0.000020... |
| 60 | 13:04:50 | SHOW CREATE TABLE cleanup_log | 1 row(s) returned | 0.061 sec / 0.000034... |
| 61 | 13:05:44 | SHOW CREATE TABLE media_retention_log | 1 row(s) returned | 0.057 sec / 0.000023... |
| 62 | 13:06:22 | SHOW CREATE TABLE auto_delete_logs | 1 row(s) returned | 0.056 sec / 0.000028... |
| 63 | 13:12:11 | SHOW PROCEDURE STATUS WHERE Db = 'dc' | 149 row(s) returned | 0.142 sec / 0.0021 sec |
| 64 | 13:15:21 | SHOW CREATE PROCEDURE dc.MediaRetentionCleanup | 1 row(s) returned | 0.061 sec / 0.000017... |
| 65 | 13:17:24 | SHOW CREATE PROCEDURE admin_del_radio | 1 row(s) returned | 0.065 sec / 0.000013... |
| 66 | 13:20:19 | SHOW CREATE PROCEDURE cleanup | 1 row(s) returned | 0.062 sec / 0.000026... |
| 67 | 13:20:59 | SHOW CREATE PROCEDURE Insert_TvVTT | 1 row(s) returned | 0.056 sec / 0.00003... |
| 68 | 13:21:22 | SHOW CREATE PROCEDURE Insert_Radio | 1 row(s) returned | 0.058 sec / 0.000011... |
| 69 | 13:33:04 | SHOW CREATE PROCEDURE Insert_Radio | 1 row(s) returned | 0.063 sec / 0.00004... |
| 70 | 13:33:19 | SHOW PROCEDURE STATUS WHERE Db = 'dc' | 149 row(s) returned | 0.140 sec / 0.00017 s... |

```
1 SHOW PROCEDURE STATUS WHERE Db = 'dc'
2
```

100% 1:1

Result Grid Filter Rows: Search Export:

| Db | Name | Type | Definer | Modified | Created | Security_ty... | Comment | character_set_cli... | collation_connecti... | Database Collation |
|---|---|---|---|---|---|---|---|---|---|---|
| dc | additional_cc | PROCEDURE | root@localhost | 2023-05-20 13:23:23 | 2023-05-20 13:23:23 | DEFINER | | utf8mb3 | utf8mb3_general_ci | utf8mb3_general_ci |
| dc | admin_active_TV | PROCEDURE | root@localhost | 2023-05-20 13:23:23 | 2023-05-20 13:23:23 | DEFINER | | utf8mb3 | utf8mb3_general_ci | utf8mb3_general_ci |
| dc | admin_delfile_srt21 | PROCEDURE | root@localhost | 2025-02-28 11:47:45 | 2025-02-28 11:47:45 | DEFINER | | utf8mb4 | utf8mb4_0900_ai_ci | utf8mb3_general_ci |
| dc | admin_del_err | PROCEDURE | root@localhost | 2023-05-20 13:23:23 | 2023-05-20 13:23:23 | DEFINER | | utf8mb3 | utf8mb3_general_ci | utf8mb3_general_ci |
| dc | admin_del_radio | PROCEDURE | root@localhost | 2023-05-20 13:23:23 | 2023-05-20 13:23:23 | DEFINER | | utf8mb3 | utf8mb3_general_ci | utf8mb3_general_ci |
| dc | admin_del_srt | PROCEDURE | root@localhost | 2023-05-20 13:23:23 | 2023-05-20 13:23:23 | DEFINER | | utf8mb3 | utf8mb3_general_ci | utf8mb3_general_ci |
| dc | admin_del_srt21 | PROCEDURE | root@localhost | 2023-05-20 13:23:23 | 2023-05-20 13:23:23 | DEFINER | | utf8mb4 | utf8mb4_0900_ai_ci | utf8mb3_general_ci |
| dc | admin_del_srt21b | PROCEDURE | root@localhost | 2023-05-20 13:23:23 | 2023-05-20 13:23:23 | DEFINER | | utf8mb4 | utf8mb4_0900_ai_ci | utf8mb3_general_ci |
| dc | Bool_Search | PROCEDURE | root@localhost | 2023-05-20 13:23:23 | 2023-05-20 13:23:23 | DEFINER | | utf8mb3 | utf8mb3_general_ci | utf8mb3_general_ci |
| dc | Checkin_Historic | PROCEDURE | localhost@root | 2023-05-20 13:23:23 | 2023-05-20 13:23:23 | DEFINER | | utf8mb4 | utf8mb4_0900_ai_ci | utf8mb3_general_ci |
| dc | cleanup | PROCEDURE | henry_VPN@... | 2025-10-27 19:24:47 | 2025-10-27 19:24:47 | DEFINER | | utf8mb4 | utf8mb4_0900_ai_ci | utf8mb3_general_ci |
| dc | CleanupOldSRT | PROCEDURE | henry_VPN@... | 2026-02-09 14:55:58 | 2026-02-09 14:55:58 | DEFINER | | utf8mb4 | utf8mb4_0900_ai_ci | utf8mb3_general_ci |
| dc | DeleteTranslation | PROCEDURE | root@localhost | 2025-12-03 18:19:46 | 2025-12-03 18:19:46 | DEFINER | | utf8mb4 | utf8mb4_0900_ai_ci | utf8mb3_general_ci |
| dc | Delete_Admin | PROCEDURE | root@localhost | 2025-05-30 03:09:09 | 2025-05-30 03:09:09 | DEFINER | | utf8mb4 | utf8mb4_0900_ai_ci | utf8mb3_general_ci |
| dc | Delete_AdminRole | PROCEDURE | root@localhost | 2025-05-30 03:08:05 | 2025-05-30 03:08:05 | DEFINER | | utf8mb4 | utf8mb4_0900_ai_ci | utf8mb3_general_ci |
| dc | Delete_AlertID_Fr... | PROCEDURE | root@localhost | 2023-05-20 13:23:23 | 2023-05-20 13:23:23 | DEFINER | | utf8mb3 | utf8mb3_general_ci | utf8mb3_general_ci |
| dc | Delete_TVConfig | PROCEDURE | henry_VPN@... | 2025-06-29 20:33:23 | 2025-06-29 20:33:23 | DEFINER | | utf8mb4 | utf8mb4_0900_ai_ci | utf8mb3_general_ci |
| dc | Delete_TVConfig... | PROCEDURE | henry_VPN@... | 2025-07-01 20:16:54 | 2025-07-01 20:16:54 | DEFINER | | utf8mb4 | utf8mb4_0900_ai_ci | utf8mb3_general_ci |
| dc | FetchAdapterError | PROCEDURE | root@localhost | 2023-05-20 13:23:23 | 2023-05-20 13:23:23 | DEFINER | | utf8mb4 | utf8mb4_0900_ai_ci | utf8mb3_general_ci |
| dc | fetchAllErrorData | PROCEDURE | root@localhost | 2025-02-06 15:47:59 | 2025-02-06 15:47:59 | DEFINER | | utf8mb4 | utf8mb4_0900_ai_ci | utf8mb3_general_ci |
| dc | FetchErrorFreque... | PROCEDURE | root@localhost | 2023-05-20 13:23:23 | 2023-05-20 13:23:23 | DEFINER | | utf8mb4 | utf8mb4_0900_ai_ci | utf8mb3_general_ci |
| dc | FetchErrorYearFreq | PROCEDURE | root@localhost | 2024-11-19 10:00:29 | 2024-11-19 10:00:29 | DEFINER | | utf8mb4 | utf8mb4_0900_ai_ci | utf8mb3_general_ci |
| dc | FetchErrorYearrFreq | PROCEDURE | root@localhost | 2023-05-20 13:23:23 | 2023-05-20 13:23:23 | DEFINER | | utf8mb4 | utf8mb4_0900_ai_ci | utf8mb3_general_ci |
| dc | FetchErrorYrMnW... | PROCEDURE | root@localhost | 2023-05-20 13:23:23 | 2023-05-20 13:23:23 | DEFINER | | utf8mb4 | utf8mb4_0900_ai_ci | utf8mb3_general_ci |
| dc | FetchffmpegAdapt... | PROCEDURE | root@localhost | 2023-05-20 13:23:23 | 2023-05-20 13:23:23 | DEFINER | | utf8mb4 | utf8mb4_0900_ai_ci | utf8mb3_general_ci |
| dc | FetchffmpegError... | PROCEDURE | root@localhost | 2023-05-20 13:23:23 | 2023-05-20 13:23:23 | DEFINER | | utf8mb4 | utf8mb4_0900_ai_ci | utf8mb3_general_ci |
| dc | FetchffmpegFrequ... | PROCEDURE | root@localhost | 2023-05-20 13:23:23 | 2023-05-20 13:23:23 | DEFINER | | utf8mb4 | utf8mb4_0900_ai_ci | utf8mb3_general_ci |
| dc | FetchffmpegStatio... | PROCEDURE | root@localhost | 2023-05-20 13:23:23 | 2023-05-20 13:23:23 | DEFINER | | utf8mb4 | utf8mb4_0900_ai_ci | utf8mb3_general_ci |

Result Grid
Form Editor
Field Types
Query Stats
Execution Plan

# Cleanup Procedures – TvVTT, RadioClips, and SRT Tables

**Overview**
After implementing the timestamp migration and introducing the indexed TStamp_dt column, cleanup procedures were created to automatically remove media records older than 90 days. These procedures provide a controlled and efficient retention mechanism for transcript and media tables within the DigiClips database.

**What This Change Does**
Creates stored procedures to delete records older than 90 days from the TvVTT, RadioClips, SRT, SRT21, and SRT21b tables.
Uses the indexed TStamp_dt column for TvVTT and RadioClips to perform efficient time-based filtering.
Implements batch deletions using LIMIT to prevent long table locks and reduce performance impact on large datasets.
Tracks cleanup activity by inserting logs into the auto_delete_logs table, including the data type cleaned, number of deleted records, execution time, and cleanup status.

**Why This Was Needed**
Previously, retention logic relied on VARCHAR timestamps or function-based filtering using STR_TO_DATE(). These approaches prevent MySQL from using indexes efficiently and can lead to slow queries and full table scans.

The new cleanup procedures improve performance by using indexed DATETIME columns for comparisons, running deletions in controlled batches, providing visibility into cleanup activity through logging, and standardizing retention logic across multiple transcript tables. This approach improves database scalability and ensures that older transcript records are automatically removed according to the 90-day retention policy.

Batch Deletion Strategy
To avoid locking large tables for extended periods, cleanup procedures delete records in batches using the LIMIT clause.

Example:
```
DELETE FROM TvVTT
WHERE TStamp_dt < DATE_SUB(NOW(), INTERVAL 90 DAY)
LIMIT 10000;
```

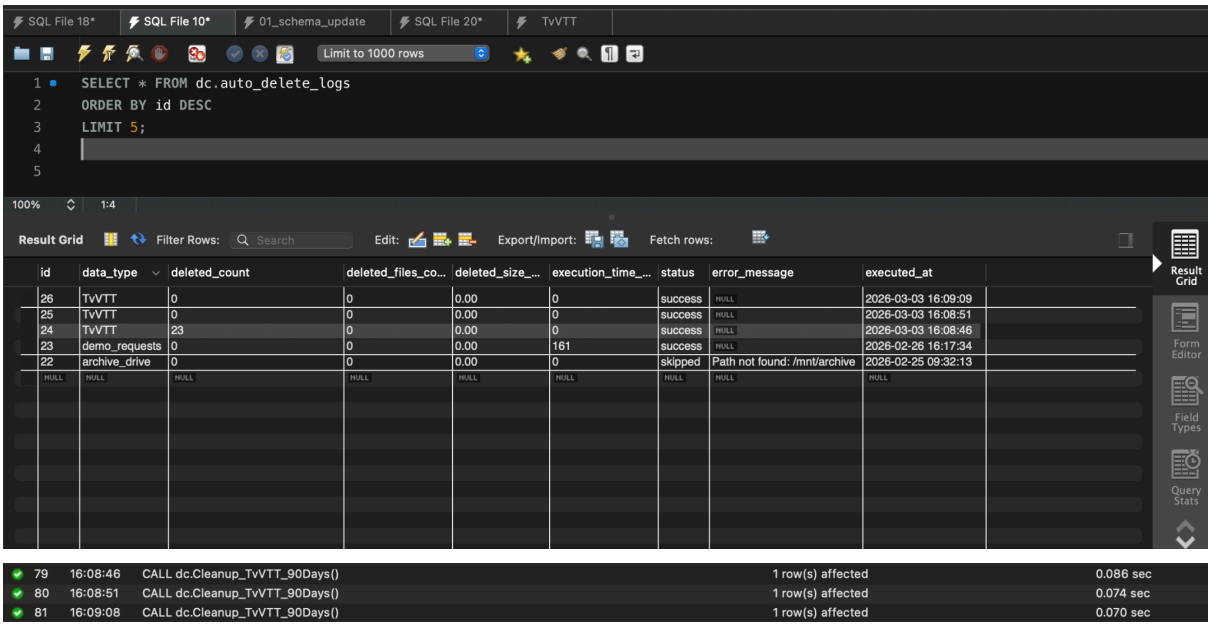
If more records remain beyond the limit, the procedure can be executed again to continue the cleanup process.

**Cleanup Logging**
Each cleanup operation records the result in the existing auto_delete_logs table. The log captures the table cleaned, number of records deleted, execution time in milliseconds, and cleanup status. This allows monitoring of retention operations and helps track maintenance activity in the database.

Verification Queries
Check if records older than 90 days still exist:

```
SELECT COUNT(*)
FROM TvVTT
WHERE TStamp_dt < DATE_SUB(NOW(), INTERVAL 90 DAY);
```

Verify cleanup log entries:

```
SELECT *
FROM auto_delete_logs
ORDER BY id DESC
LIMIT 20;
```

These queries confirm that the cleanup procedures executed correctly and that the retention policy is functioning as expected.